

Task #1: Backtracking for Sudoku

1) State-space tree for:

1		3
	1	
2		

State-space tree

Hint: Answer is

1	2	3
3	1	2
2	3	1

2) State-space tree for:

1			4
	4	1	
	1	4	
4			1

State-space tree

Hint: Answer is

1	2	3	4
3	4	1	2
2	1	4	3
4	3	2	1

3) Apply State-space tree to solve:

	2		
		3	4
3	4		
		2	

Your answer is

	2		
		3	4
3	4		
		2	

4) Apply State-space tree to solve:

	1		4
		2	
1			3
	4		

Your answer is

	1		4
		2	
1			3
	4		

Task #2: To use Branch & Bound to solve Assignment Problems

1)

	Job 1	Job 2	Job 3	Job 4
Person A	2	9	7	8
Person B	4	6	3	7
Person C	7	5	1	8
Person D	6	7	9	4

2)

	Job 1	Job 2	Job 3	Job 4
Person A	2	9	7	1
Person B	4	6	3	2
Person C	7	5	2	1
Person D	5	3	1	4